Checkpoint 6 Maths — Practice Test 2

Paper 1: 45 minutes · no calculator · Paper 2: 45 minutes · calculator allowed · Total 80 marks

Answer all questions. Write your answers in the spaces provided. Show your working where there is space for it. The number of marks for each question is shown in brackets [ ].

Name _____ Class _____ Date _____ Mark: ____ / 80

Abdelkriem Academy · practice paper written in the style of Cambridge Checkpoint. Not an official Cambridge paper.

Paper 1 — 45 minutes — 40 marks

1 **[1]**

What is the value of the digit **7** in the number **3.472**?

- ☐ A) 7 tenths
- ☐ B) 7 hundredths
- ☐ C) 7 thousandths
- ☐ D) 7 ones

2 The freezer in a supermarket is at −18 °C. The fridge next to it is 22 degrees warmer. **[1]**

What is the temperature in the fridge?

Working

Answer _____ °C

3 The numbers in this sequence go down by the same amount each time. **[1]**

1.4, 1.0, 0.6, 0.2, …

Write the next number in the sequence.

Working

Answer _____

4 **[1]**

Write the missing number.

$4.6 \times \square = 4600$

Working

Answer

5 **[1]**

Write the missing number.

$5.208 = 5 + 0.2 + \square$

Working

Answer

6 **[1]**

Work out.

$36 - 4 \times 5 + 2$

Working

Answer

7 Here is a calculation. It is not correct yet. **[1]**

$6 + 4 \times 5 - 2 = 30$

Write brackets in the calculation to make it correct. Write the whole calculation on paper.

8 A cinema sells 1375 tickets every week. **[2]**

How many tickets does it sell in 24 weeks? Show your working.

9 456 people are travelling to a festival by bus. Each bus can carry 32 people. **[2]**

How many buses are needed? Show your working.

10 Bus route 12 leaves the bus station every 6 minutes. Bus route 30 leaves every 8 minutes. Both buses leave together at 07:00. **[1]**

At what time do both buses next leave together?

11 Omar says, "2016 is divisible by 9." **[1]**

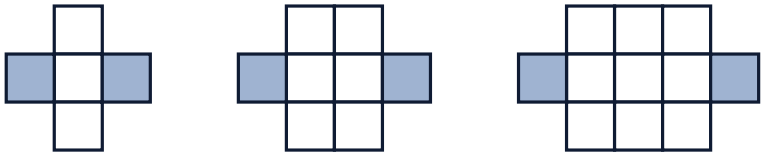
Is Omar correct? Explain how you know without doing the division.

12 **[1]**

Which number is both a **square number** and a **cube number**?

- ☐ A) 8
- ☐ B) 16
- ☐ C) 27
- ☐ D) 64

13 Laila makes patterns with white tiles and grey tiles. Each new pattern has 3 more white tiles than the one before. **[1]**



Pattern 1 Pattern 2 Pattern 3

(a) How many tiles (white and grey) are in **Pattern 10**?

Working

Answer

(b) Which rule gives the number of tiles in **any** pattern? **[1]**

- ☐ A) number of tiles = 3 × position + 2
- ☐ B) number of tiles = position + 3
- ☐ C) number of tiles = 2 × position + 3
- ☐ D) number of tiles = 5 × position

(c) Can one of Laila's patterns have exactly **40** tiles? Explain how you know. **[1]**

14 The rule for a sequence is: **[1]**
term = position × position + 3

What is the **5th** term of the sequence?

Working

Answer

15 At the supermarket, a bag of oranges costs **p** pounds. A bag of apples costs 15 pounds more than a bag of oranges. **[1]**

Write an expression for the cost of a bag of apples, in pounds.

16 **[1]**

Which fraction is 18/24 written in its **simplest form**?

- ☐ A) 9/12
- ☐ B) 6/8
- ☐ C) 3/4
- ☐ D) 2/3

17 **[1]**

Write **17 ÷ 5** as a mixed number.

18 **[1]**

Work out 3/5 of 45.

Working

Answer

19 Salma has 2 1/4 kg of rice. She uses 5/6 kg of it to make koshari. **[2]**

How much rice does she have left? Give your answer as a mixed number. Show your working.

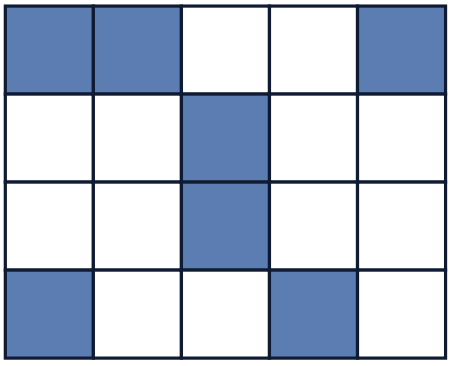
Working

Answer kg

20 **[1]**

Work out 4/5 ÷ 2. Give your answer as a fraction.

21 Here is a tile design made of equal squares. **[1]**



What **percentage** of the design is shaded?

Working

Answer ____ %

22 Hana says, "0.45 is bigger than 0.5 because 45 is bigger than 5." **[1]**

Explain why Hana is **wrong**.

23 At the supermarket, Karim buys a bag of rice for E£34.75 and a packet of tea for E£18.50. He pays with a E£100 note. **[2]**

How much change should he get? Show your working.

Working

Answer ____ E£

24 One floor tile costs E£4.35. **[1]**

How much do 6 tiles cost?

Working

Answer E£

25 **[1]**

Round **7.46** to the nearest tenth.

Working

Answer

26 A film at the cinema lasts 1 hour 45 minutes. **[1]**

Write 1 hour 45 minutes as a decimal number of hours.

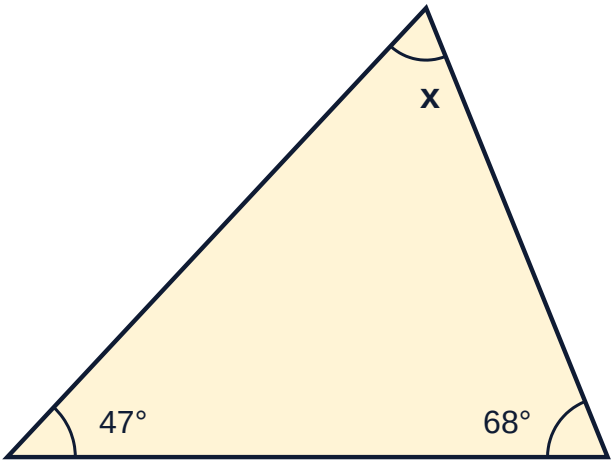
Working

Answer hours

27 **[1]**

Sketch a quadrilateral that has **exactly one pair of parallel sides** and **exactly two right angles**. Draw it on paper.

28 Here is a triangle-shaped tile. **[1]**

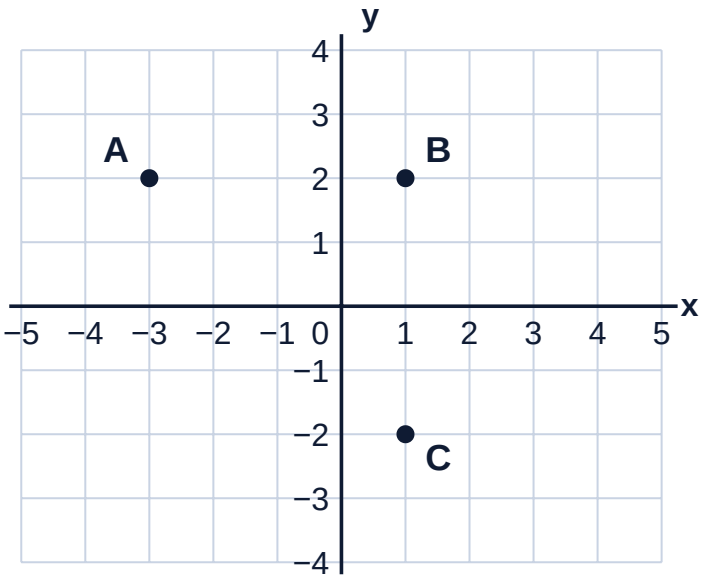


Work out the size of angle **x**.

Working

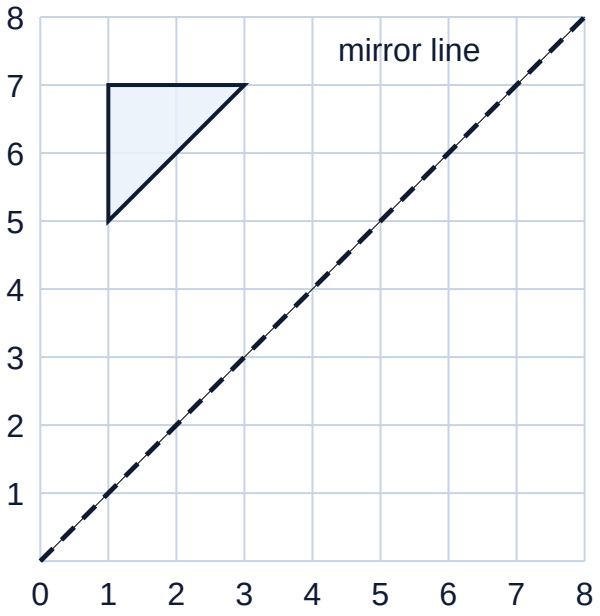
Answer ____ °

29 A, B and C are three corners of a square. D is the fourth corner. **[1]**



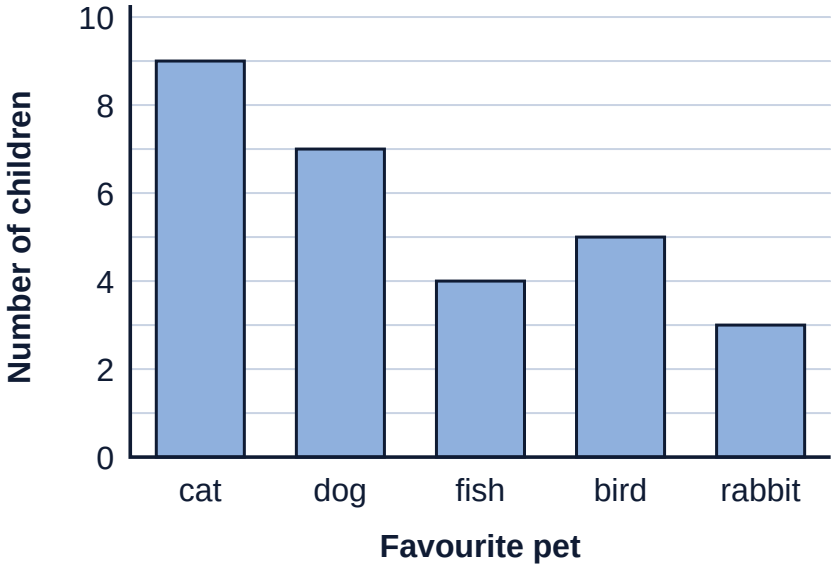
Write the coordinates of point **D**.

30 Here is a triangle on a square grid and a diagonal mirror line. **[1]**



Copy the grid and draw the reflection of the triangle in the mirror line.

31 Every child in Class 6B said their favourite pet. The bar chart shows the results. **[1]**



(a) How many children are in Class 6B?

Working

Answer

(b) Which favourite pet is the **mode**? **[1]**

(c) Hamza says, "More than half of the class chose a cat or a dog." Is Hamza correct? Show how you know. **[1]**

32 In a game, a bag has 6 cat stickers, 3 dog stickers and 1 fish sticker. Rania takes one sticker without looking. **[1]**

Which word best describes the chance that Rania takes a cat sticker **or** a dog sticker?

- ☐ A) impossible
- ☐ B) unlikely
- ☐ C) even chance
- ☐ D) likely

Paper 2 — 45 minutes — 40 marks

1 A bus leaves Saint Catherine early in the morning when the temperature is −3 °C. It arrives in Sharm El-Sheikh, where the temperature is 24 °C. **[1]**

By how many degrees is Sharm El-Sheikh warmer?

Working

Answer ____ °C

2 A supermarket sells 2485 bottles of water each day. **[1]**

How many bottles does it sell in 28 days?

Working

Answer ____

3 Nour has 36 red tiles and 48 blue tiles. She puts them into packs. Every pack must be the same, and she must use all the tiles. **[1]**

What is the **greatest** number of packs she can make?

Working

Answer ____

4 **[1]**

Which of these numbers is divisible by **6**?

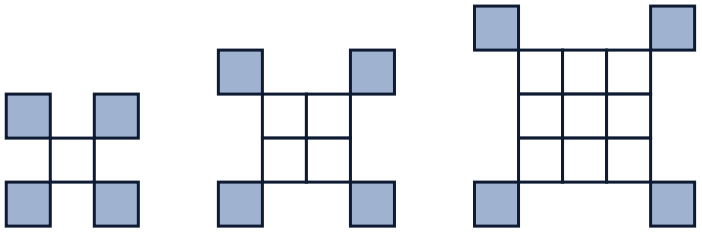
- ☐ A) 334
- ☐ B) 405
- ☐ C) 552
- ☐ D) 718

5 **[1]**

Three of these calculations give the same answer. Which one gives a **different** answer?

- ☐ A) 25 × 4 × 37
- ☐ B) 37 × 25 × 4
- ☐ C) 100 × 37
- ☐ D) 25 + 4 × 37

6 Ali makes tile patterns. Pattern n has n × n white tiles in a square, and 4 grey tiles at the corners. **[1]**



Pattern 1 Pattern 2 Pattern 3

How many tiles (white and grey) are in **Pattern 12**?

Working

Answer

7 Here is part of a cinema timetable. **[1]**

Film	Starts	Length
Desert Falcon	14:35	1 hour 50 minutes
Space Cats	15:10	1.4 hours
The Lost Pyramid	16:45	2.25 hours

(a) At what time does *Desert Falcon* end?

(b) *Space Cats* lasts 1.4 hours. Complete: 1.4 hours = 1 hour and ☐ minutes. **[1]**

Working

Answer minutes

(c) Hany watches *Desert Falcon*. After the film he walks to the bus stop, which takes 20 minutes. His bus leaves at 16:40. Will he catch the bus? Show how you know. **[1]**

8 Mariam travels to school by bus. **[1]**

(a) A card for 12 bus journeys costs E£81.60. How much does one journey cost with the card?

Working

Answer E£

(b) The bus travels at a steady speed. It travels 18 km in 24 minutes. How far does it travel in 40 minutes? Show your working. **[2]**

9 A supermarket has a sale. **[1]**

(a) A 2 kg bag of rice usually costs E£64. In the sale, the price is reduced by 15%. What is the sale price?

Working

Answer E£

(b) Pack A has 6 yogurts for E£37.20. Pack B has 8 yogurts for E£47.60. Which pack is better value? Show your working. **[2]**

(c) A can of beans costs E£23.75. How much do 12 cans cost? **[1]**

Working

Answer E£

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10 **[1]**

Which list is in order from **smallest to largest**?

- ☐ A) 0.69, 0.7, 72%, 3/4
- ☐ B) 0.7, 0.69, 72%, 3/4
- ☐ C) 0.69, 72%, 0.7, 3/4
- ☐ D) 3/4, 72%, 0.7, 0.69

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11 Here is a recipe for koshari for **6 people**. **[1]**

Ingredient	Amount
rice	360 g
brown lentils	240 g
macaroni	300 g
tomato sauce	450 ml
onions	3

(a) How much rice is needed to make koshari for **10 people**?

Working

Answer g

(b) Write the ratio **rice : lentils** in its simplest form. **[1]**

(c) Karim has a 600 ml bottle of tomato sauce. He uses 3/4 of the bottle. How many millilitres does he use? **[1]**

Working

Answer ml

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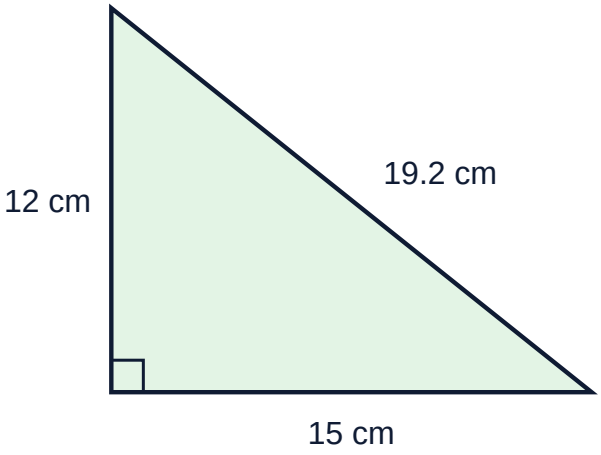
12 Mona cooks koshari in a large pot. **[1]**

Which sentence is about the **capacity** of the pot?

- ☐ A) The metal of the pot takes up 600 cm^3 of space.
- ☐ B) The pot can hold 5 litres of food.
- ☐ C) The pot has a mass of 1.2 kg.
- ☐ D) The base of the pot has an area of 490 cm^2.

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13 Nour designs tiles for a kitchen wall. **[1]**

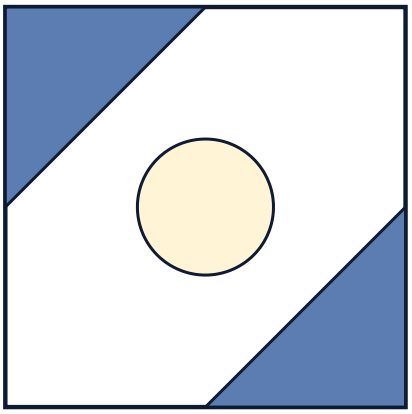


(a) One tile is a right-angled triangle. Work out its area.

Working

Answer cm^2

[1]



(b) Here is another of Nour's tiles. What is its order of rotational symmetry?

- ☐ A) order 1
- ☐ B) order 2
- ☐ C) order 3
- ☐ D) order 4

(c) Nour covers **all** the faces of a cube-shaped box with square tiles. The edges of the box are 20 cm long. Each tile is 10 cm by 10 cm. How many tiles does she need? **[1]**

Working

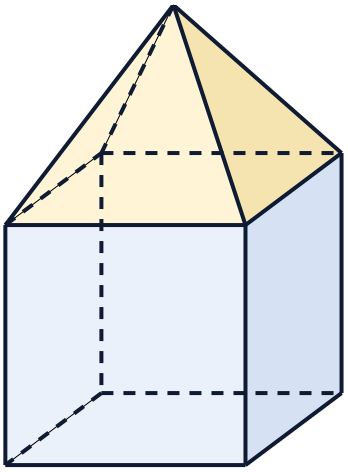
Answer

(d) Nour says, "Every square is also a rectangle." Is she correct? Explain your answer. **[1]**

14 **[1]**

Sketch a net of a **triangular prism**. Draw it on paper.

15 This shape is made from a cube with a square-based pyramid on top. **[1]**

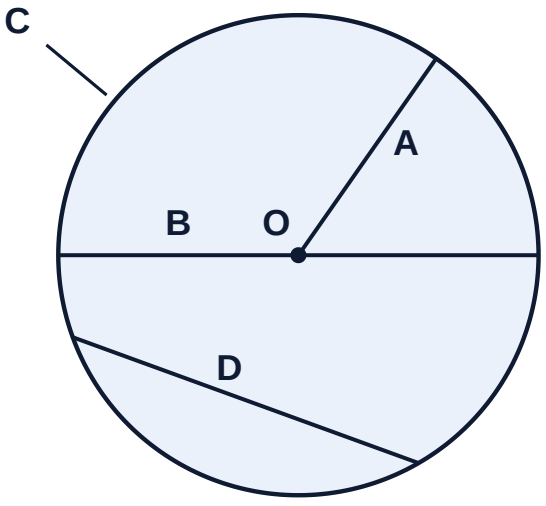


How many **faces** does the shape have?

Working

Answer

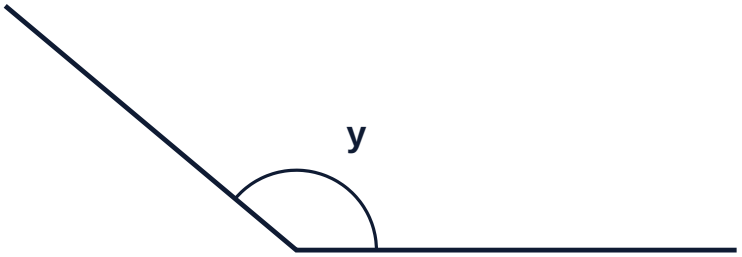
16 Here is a circle with centre O. **[1]**



(a) Which letter shows a **radius** of the circle?

(b) Use compasses to draw a circle with a **diameter of 8 cm** on paper. Mark its centre. **[1]**

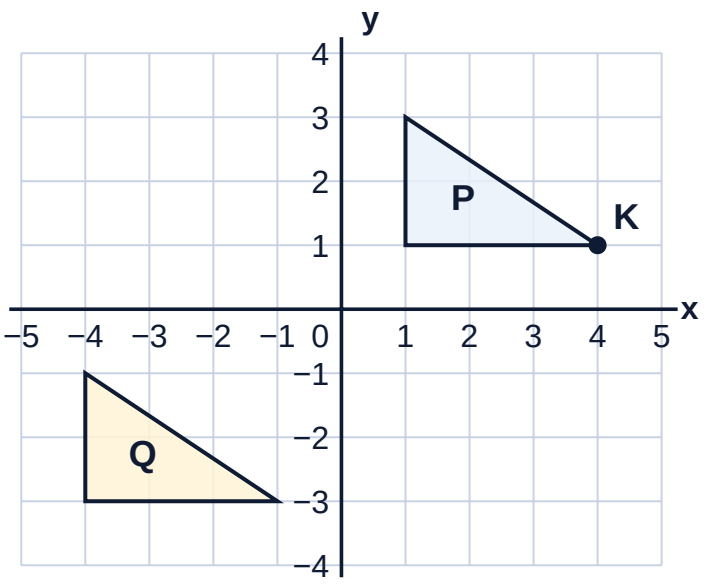
17 **[1]**



Which is the best **estimate** of the size of angle **y**?

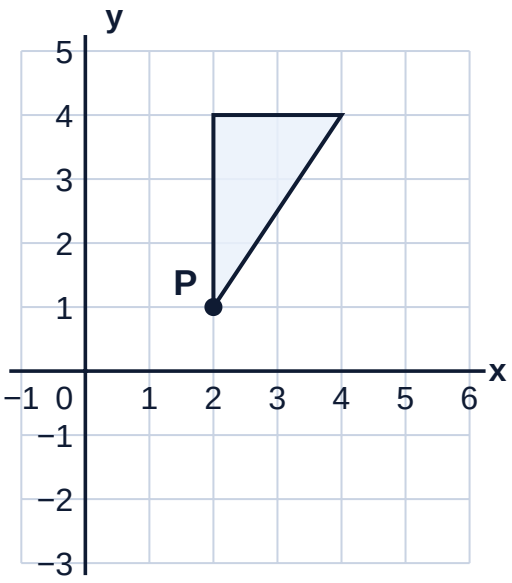
- ☐ A) 40°
- ☐ B) 90°
- ☐ C) 140°
- ☐ D) 220°

18 Triangle P is translated to triangle Q. **[1]**



Point K on triangle P moves to a corner of triangle Q. Write the coordinates of this corner.

19 Here is a triangle on a coordinate grid. **[1]**



Copy the grid and rotate the triangle **90° clockwise** about the vertex **P**. Draw the new triangle.

20 Ten children wrote down how many pets they have at home. **[1]**

2, 0, 1, 16, 1, 3, 0, 4, 1, 2

(Omar has 16 fish in a fish tank.)

(a) Work out the **mean** number of pets.

Working

Answer

(b) Work out the **median** number of pets. **[1]**

Working

Answer

(c) Which average, the mean or the median, better describes the number of pets a typical child has? Give a reason. **[1]**

21 Class 6B plans an investigation about pets. **[1]**

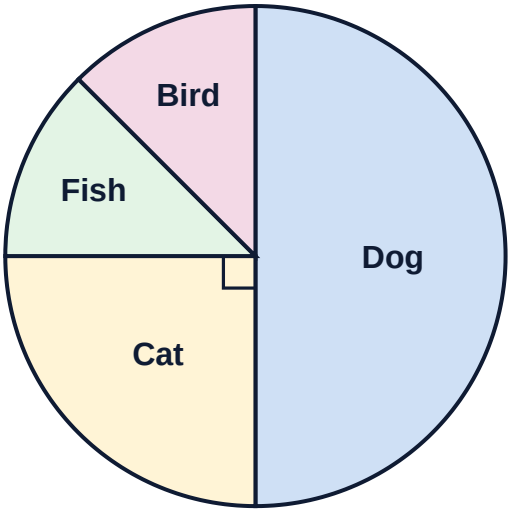
Which of these is **continuous** data?

- ☐ A) the mass of each pet in kilograms
- ☐ B) the number of pets each child has
- ☐ C) the type of each pet
- ☐ D) the number of legs each pet has

22 Salma weighs her puppy every week for 8 weeks to see how its mass changes. **[1]**

Which would be the best way to show her results: a **Venn diagram**, a **bar chart** or a **line graph**? Explain your choice.

23 40 children were asked about their favourite pet. The pie chart shows the results. **[1]**



How many children chose **cat**?

Working

Answer

24 Yara rolls an ordinary dice numbered 1 to 6 once. **[1]**

Which two events are **mutually exclusive** (cannot happen at the same time)?

- ☐ A) getting an even number and getting a 4
- ☐ B) getting a number less than 3 and getting an odd number
- ☐ C) getting a 6 and getting an odd number
- ☐ D) getting a number greater than 2 and getting an even number

25 Nadia spins a spinner 20 times and gets red 13 times. Karim spins the same spinner 500 times and gets red 260 times. **[1]**

Whose results give a better estimate of the probability of getting red? Explain why.